

TP — Browser Fingerprinting

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Alice deleted her cookies and blocked storage. The question is whether a server can still recognise her browser.

I built a small Flask site that collects three kinds of information: the HTTP headers, some browser properties read with JavaScript, and the way the user types. I tested it with Chrome, Chrome Incognito, Safari and Brave on the same MacBook.

1 Part 1: passive fingerprinting

The browser sends headers automatically with every request. The server does not need JavaScript and does not ask for anything. It only reads what already arrived.

Header	What it shows	Useful?
User-Agent	Browser, version, operating system	Yes
Accept-Language	Preferred languages, in order	Yes
Accept	Content types the browser prefers	A little
Accept-Encoding	Compression formats supported	A little
Sec-CH-UA	Browser brand (Chromium only)	Yes
Sec-CH-UA-Platform	Operating system	Yes
Sec-CH-UA-Mobile	Phone or desktop	Barely
Sec-Fetch-*	How the user reached the page	No
DNT, Sec-GPC	Privacy signals, if enabled	Rare, so yes
Referer	The previous page	No

Table 1: The headers I collected and what they reveal.

```
===== NEW VISIT =====
IP address : 127.0.0.1
HTTP method : GET
User-Agent   : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : fr-FR,fr;q=0.9,en-US;q=0.8,en;q=0.7
Accept       : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding : gzip, deflate, br, zstd
Sec-CH-UA    : "Chromium";v="154", "Google Chrome";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Mobile : 70
Sec-CH-UA-Platform : "macOS"
Sec-Fetch-Site : none
Sec-Fetch-Mode : navigate
Sec-Fetch-Dest : document
Upgrade-Insecure-Requests : 1
DNT            : None
Sec-GPC        : None
Referer        : None
Connection     : keep-alive

127.0.0.1 - - [29/Sep/2026 11:27:59] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 11:27:59] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
```

Figure 1: The Flask terminal after one visit. All the headers arrive without JavaScript and without cookies.

What changes and what does not. I reloaded the page in a normal window, in a private window and in another browser.

Chrome and Chrome Incognito gave exactly the same headers. Nothing changed at all. Private mode clears cookies and history, but it does not change what the browser is.

Safari changed the `User-Agent` and the `Accept` header, and it did not send the `Sec-CH-UA` headers at all. On the other hand `Accept-Language` stayed the same, because it comes from the system settings and not from the browser. The IP address also stayed the same.

These headers show the browser family and the language, but millions of people share the same combination. They are a starting point, not an identity.

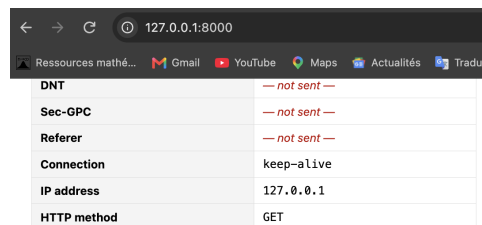
2 Part 2: active fingerprinting

Here the server sends a JavaScript file. The script asks the browser questions that the HTTP request never answers.

Feature	Object	What it shows
language, languages	navigator	The full list of languages, in order
userAgent	navigator	Browser, version, system
platform	navigator	The architecture (MacIntel)
hardwareConcurrency	navigator	Number of CPU threads (8 here). Hardware
deviceMemory	navigator	RAM in GB (16). Chromium browsers only
screenResolution	screen	Size of the screen (1470 × 956)
availableScreen	screen	Screen minus the menu bar and the Dock
colorDepth	screen	Bits per pixel. Chrome says 30, which is rare
windowSize	window	The window. Changes when I resize it
devicePixelRatio	window	Retina scaling and zoom level
timeZone	Intl	Africa/Casablanca
locale	Intl	Language and country together
referrer	document	The previous page

Table 2: The features collected by JavaScript.

The useful ones are the hardware features. The number of CPU threads, the screen resolution and the time zone describe the machine, so they do not depend on which browser I open. The window size is the opposite case: it has many possible values, but it changes as soon as I resize the window.



DNT	— not sent —
Sec-GPC	— not sent —
Referer	— not sent —
Connection	keep-alive
IP address	127.0.0.1
HTTP method	GET

Part 2 — Active Feature Collection

Feature	Value read by JavaScript
language	fr-FR
languages	fr-FR, fr, en-US, en
userAgent	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
platform	MacIntel
hardwareConcurrency	8
deviceMemory	16
screenResolution	1470x956
availableScreen	1470x924
colorDepth	30
windowSize	624x885
devicePixelRatio	1.7999999523162842
timeZone	Africa/Casablanca
locale	en-US
referrer	(none)

Figure 2: The features displayed on the page.

```

----- ACTIVE FEATURES -----
language       : fr-FR
languages      : fr-FR, fr, en-US, en
userAgent      : Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
platform       : MacIntel
hardwareConcurrency : 8
deviceMemory   : 16
screenResolution : 1470x956
availableScreen : 1470x924
colorDepth     : 30
windowSize     : 624x885
devicePixelRatio : 1.7999999523162842
timeZone       : Africa/Casablanca
locale         : en-US
referrer       : (none)
-----

```

Figure 3: The same features arrive on the server after the `fetch()` call.

Part 2 — Active Feature Collection

Feature	Value read by JavaScript
language	fr-FR
languages	fr-FR, fr, en-US, en
userAgent	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
platform	MacIntel
hardwareConcurrency	8
deviceMemory	16
screenResolution	1470x956
availableScreen	1470x924
colorDepth	30

Figure 4: Developer Tools. The `collect` request leaves the browser with the features inside it.

Testing the protections

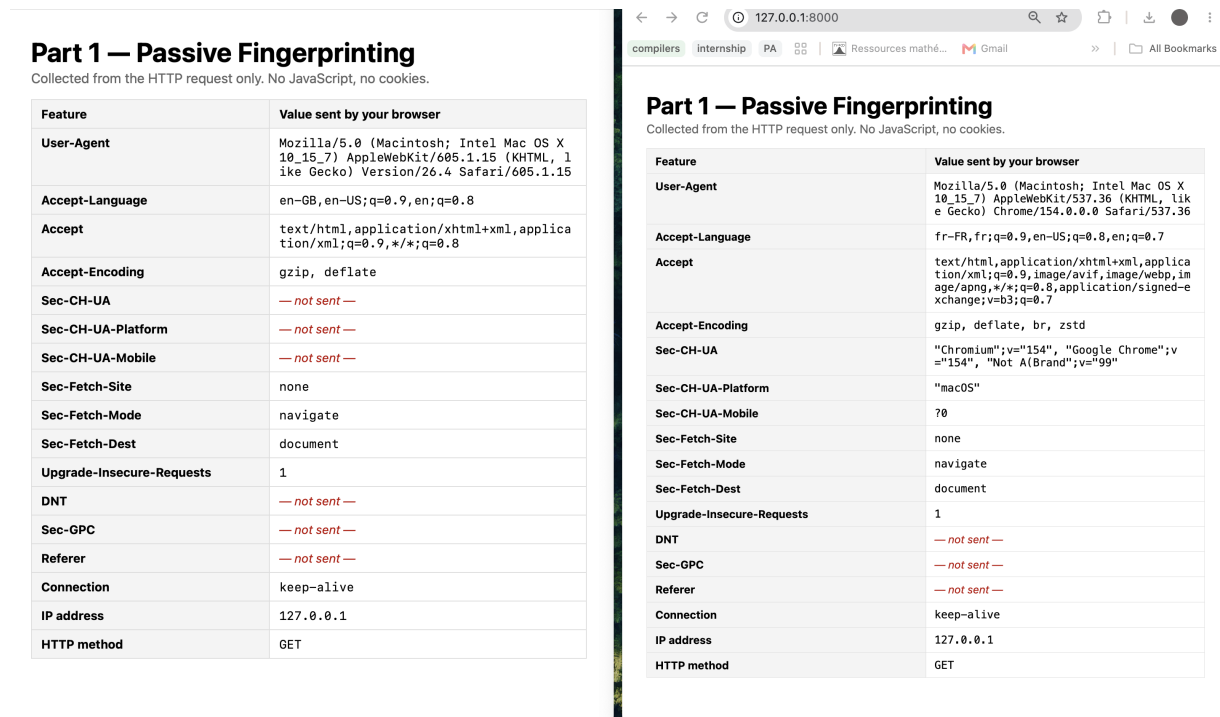


Figure 5: Safari (left) and Chrome Incognito (right) on the same MacBook.

Feature	Chrome	Safari	Why
userAgent	Chrome/154	Version/26.4	Different browser
deviceMemory	16	unavailable	Safari has no such API
colorDepth	30	24	Safari always answers 24
devicePixelRatio	1.799...	2	Chrome adds my 90% zoom
language	fr-FR	en-GB	My own setting in each browser
screenResolution	1470 × 956		Same machine
hardwareConcurrency	8		Same machine
timeZone	Africa/Casablanca		Same machine

Table 3: Chrome and Safari compared.

Changing the browser changes the browser rows, but the machine rows stay identical. A tracker that looks only at those rows can still link my two browsers together.

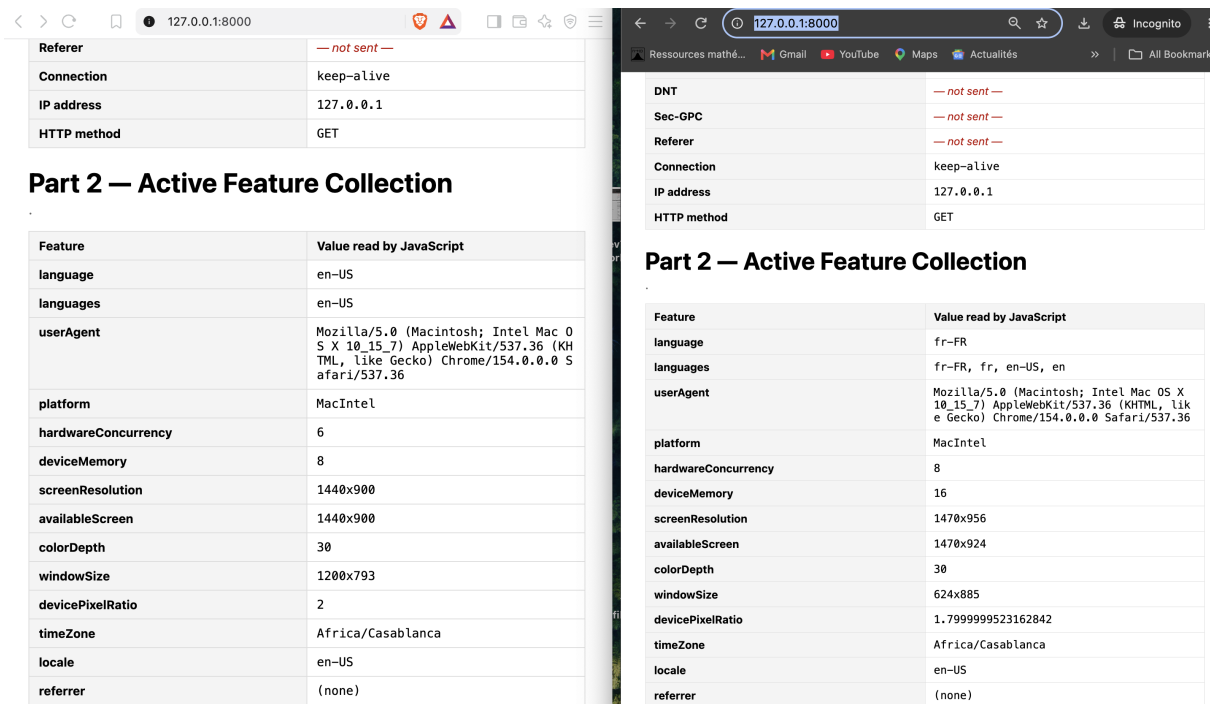


Figure 6: Brave (left) and Chrome (right) on the same MacBook. Brave does not give the real values.

Feature	Real value	Brave	Comment
hardwareConcurrency	8	6	False number
deviceMemory	16	8	False number
screenResolution	1470 × 956	1440 × 900	Rounded
availableScreen	1470 × 924	1440 × 900	Rounded
language	fr-FR	en-US	A very common value

Table 4: Brave lies about the values. This technique is called *farbling*.

So private browsing and cookie blocking protect the storage, not the identity. Brave is much better because it gives false values. The best protection is uniformity, like the Tor Browser or Firefox with the resistFingerprinting option, where every user looks the same.

None of this is perfect. A rare protection can make a user easier to recognise, not harder. Random values can be detected when they do not match each other. Some sites break. And the headers and the IP address still leak.

3 Part 3: typing behaviour

Typing speed uses the usual convention that one word is five characters:

$$\text{WPM} = \frac{\text{characters} \div 5}{\text{time in minutes}}$$

The algorithm is short:

1. Start with no start time, zero corrections, and the test not finished.
2. On keydown, save the start time at the first printable character.
3. On keydown, add one correction for Backspace or Delete.
4. On input, stop as soon as the text equals the target sentence.
5. Compute the time and the speed, display them, send them to Flask.
6. Reset everything when the Restart button is clicked.

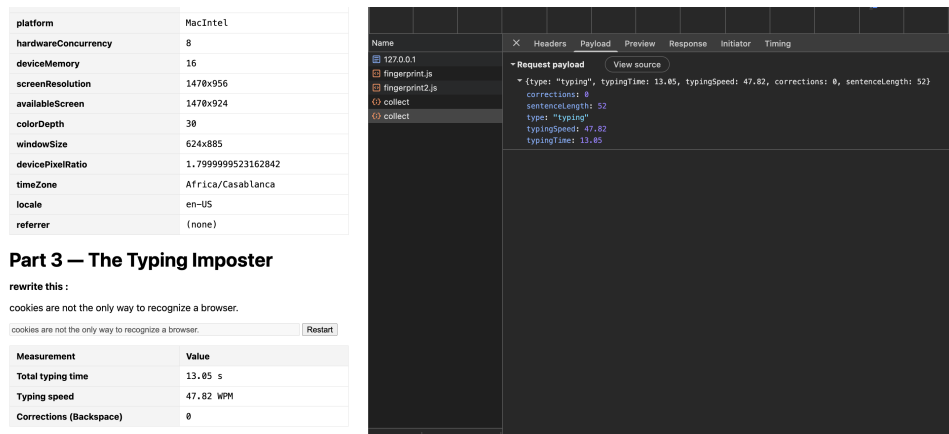


Figure 7: The second collect request. The Payload tab shows the data leaving the browser.

```
127.0.0.1 - - [29/Sep/2026 12:24:12] "POST /collect HTTP/1.1" 200 -  
----- TYPING BEHAVIOUR -----  
Total time       : 13.05 s  
Typing speed     : 47.82 WPM  
Corrections      : 0  
Sentence length  : 52 characters  
-----  
127.0.0.1 - - [29/Sep/2026 12:26:04] "POST /collect HTTP/1.1" 200 -  
[]
```

Figure 8: The same values printed by Flask.

Sentence length	52 characters
Total time	13.05 s
Typing speed	47.82 WPM
Corrections	0

Table 5: One attempt.

This measurement is not stable. My speed changed at every attempt, mostly because I learned the sentence. It also depends on the keyboard, on fatigue and on stress. One sample proves nothing, but many samples can build a profile. This feature can support a device fingerprint, not replace it.